ENERGY EFFICIENCY MODULE-HEAT EXCHANGERS

HEAT EXCHANGER PERFORMANCE ASSESSMENT

Heat Exchanger can deteriorate with time, off design operations and other interferences such as fouling, scaling, corrosion etc., can affect its performance. It is necessary to assess periodically the heat exchanger performance in order to maintain them at a high efficiency level.

The objective of performance assessment is to determine the heat exchanger duty, overall heat transfer coefficient, heat exchanger effectiveness, process/utility side pressure drop. Any deviation from the design will indicate occurence of fouling.

Heat Exchanger(Unmixed fluids)-Metadata Information

Design Data Inputs-from Test Reports

Select the Heat Exchanger Flow type:

Counterflow

No.of Shell Side passes

1

No.of Tube Side passes

2

Total Design Heat Duty(KW)

12561.00

Hot fluid side pressured drop(bar)

0.10

Cold fluid side pressured drop(bar)

0.29

Temp.range Hot fluid(°C)

74.20

Temp.range Cold fluid(°C)

60.00

Corrected LMTD(°C)

73.74

Overall Heat Transfer Coefficient U (KW/m2-K)

1.1500

Operation Data Inputs

Hot fluid flow-Inlet(kg/hr)

1827.30

Cold fluid flow-Inlet(kg/hr)

16400.00

Hot fluid temp.-Inlet(°C)

164.20

Hot fluid temp.-Outlet(°C)

100.00

Cold fluid temp.-Inlet(°C)

20.00

Cold fluid temp.-Outlet(°C)

70.00

Hot fluid pressure-Inlet(bar g)

7.00

Hot fluid pressure-Outlet(bar g)

6.60

Cold fluid pressure-Inlet(bar g)

10.00

Cold fluid pressure-Outlet(bar g)

9.20

Fluid properties

Hot fluid specific heat capacity(KJ/kg°C)

2.01000

Cold fluid specific heat capacity(KJ/kg°C)

4.00000

Latent heat of condensation of a hot condensing vapour(KJ/kg)

2068.00000

Latent heat of Vaporisation(KJ/kg)

0.00000

How do you want to provide the Heat Transfer Area (A) m^2?

- Calculate A
- Input A Manually

Enter the Heat Transfer Area (A, in m^2)

144.00

Final Heat Transfer Area (A): 144.0000 m^2

How do you want to provide the Correction Factor (F)?

- Calculate F
- Input F Manually

Effectiveness of Heat Exchanger (ϵ): `0.3995`

Heat Capacity Ratio (r): `0.0000`

Results

	Parameters	Units	Operation Data	Design Data
0	Duty(Q)	KW	1049.68	12561.00
1	Hot fluid side Pressure drop ΔPh	bar	0.40	0.10
2	Cold fluid side Pressure drop ΔPc	bar	0.80	0.29
3	Temperature Range Hot fluid	°C	64.20	74.20
4	Temperature Range Cold fluid	°C	50.00	60.00
5	Corrected LMTD	°C	80.35	73.74
6	Heat Transfer Coefficient U	KW/m2-K	0.091	1.150

Fouling Factor (R_d): `10.1528 m²K/kW`

Effectiveness (ϵ): `0.40`

AI Performance Diagnostic Summary

Generate AI Assessment Summary

- **Thermal performance:** The overall heat-transfer coefficient has fallen to 0.091 kW m^{-2} K^{-1}, only **≈ 8 % of the design value (1.150 kW m^{-2} K^{-1})**, causing the heat duty to drop to 1 050 kW (≈ 8 % of design). The corrected LMTD is slightly higher than design (80.4 °C vs 73.7 °C), indicating the temperature driving force is being maintained but the low U-value is the dominant loss factor. Effectiveness of 0.40 is far below the typical design target (≈ 0.8–0.9), confirming severe thermal degradation.
- **Hydraulic performance:** Hot-side pressure drop has increased four-fold (0.40 bar vs 0.10 bar design) and cold-side drop has risen by ~ 176 % (0.80 bar vs 0.29 bar). The elevated ΔP on both sides points to flow-channel blockage and/or increased fluid viscosity, contributing to higher pumping power and reduced mass-flow rates.
- **Fouling severity:** The calculated fouling resistance (R_f = 10.15 m^2 K kW^{-1}) is orders of magnitude larger than typical clean-tube values (≈ 0.0001–0.001 m^2 K kW^{-1}). This level of fouling alone can account for the observed ~ 90 % loss in U-value and the pressure-drop penalties.

- **Recommended actions:**

1. **Thermal cleaning (high-pressure steam or chemical cleaning)** to remove scale, deposits, and bio-fouling; re-measure U-value after cleaning to verify recovery.
2. **Mechanical inspection** (borescope or tube-sheet removal) for tube deformation, corrosion, or blockage; replace damaged tubes if necessary.
3. **Back-flushing or pigging** of the cold-side circuit to clear particulate fouling and restore design ΔP.
4. **Review and adjust operating parameters** (flow rates, inlet temperatures, water-treatment chemistry) to mitigate future fouling and keep ΔP within design limits.